

Problem

Design an analog computer to simulate the following equation to solve for $v(t)$ (assume the initial conditions are zero):

$$(d^3 v(t)/dt^3) + 3(dv(t)/dt) = 4f(t)$$

Step-by-step solution

Step 1 of 6

The differential equation for designing the analog computer is,

$$\frac{d^3 v(t)}{dt^3} + 3 \frac{dv(t)}{dt} = 4f(t)$$

Integrate the equation with respect to t .

$$\frac{d^2 v(t)}{dt^2} + 3v(t) = 4 \int f(t) dt$$

Rewrite the above equation.

$$\frac{d^2 v(t)}{dt^2} = 4 \int f(t) dt - 3v(t) \dots\dots (1)$$

Step 2 of 6

If the input to the analog computer is $\frac{d^2 v(t)}{dt^2}$, then the output of the integrator is $\frac{-dv(t)}{dt}$.

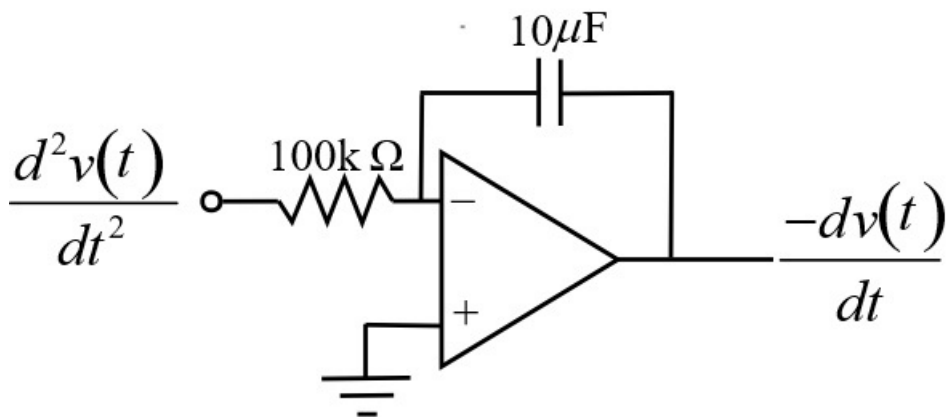


Figure 1

Step 3 of 6

If the input to the analog computer is $\frac{-dv(t)}{dt}$, then the output of the integrator is $v(t)$.

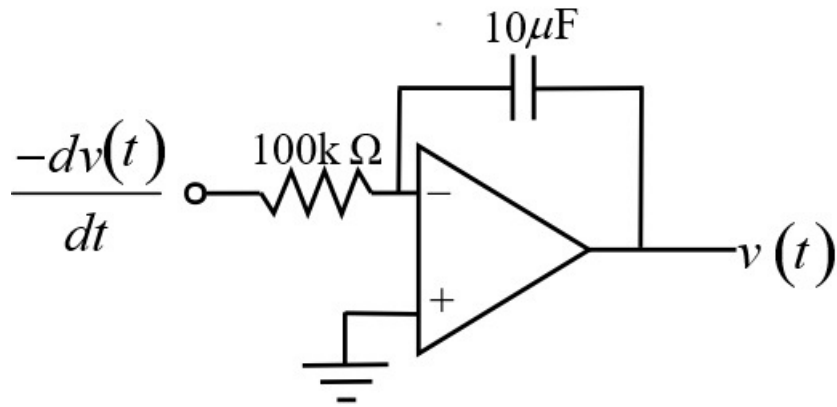


Figure 2

Step 4 of 6

Draw the circuit for $-\int f(t)dt$.

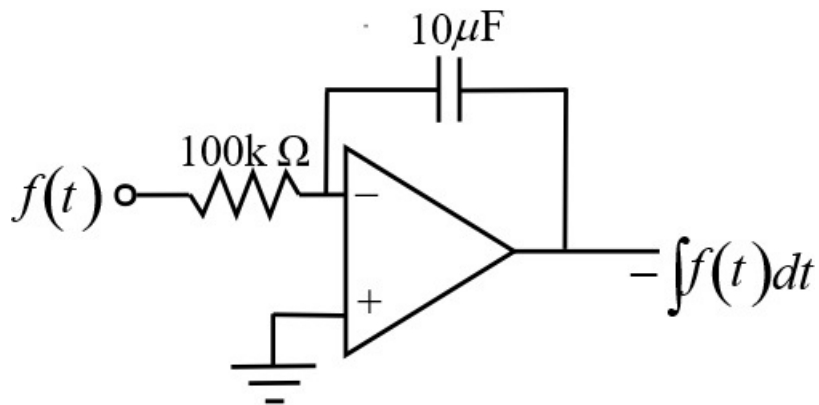
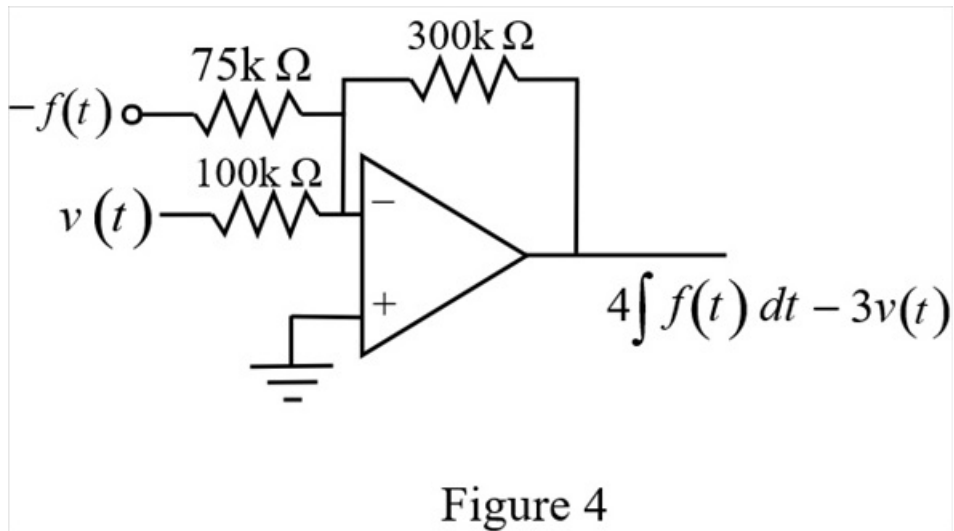


Figure 3

Step 5 of 6

Draw the inverting amplifier to make equation $4\int f(t)dt - 3v(t)$.



Step 6 of 6

Combine Figure 1, 2, 3 and 4 to draw the equation (1).

